

Architectural Studies

This section presents the requirements for programs in:

- **Design B.A.S. Honours**
- **Urbanism B.A.S. Honours**
- **Conservation and Sustainability B.A.S. Honours**

The Azrieli School of Architecture and Urbanism cooperates with the School for Studies in Art and Culture in offering the History and Theory of Architecture B.A. and B.A. Honours programs (see the Art History program section of this Calendar for details).

Course Categories for B.A.S. Programs

Core Courses

Students in B.A.S. programs must refer to the calendar which corresponds to their program catalog year to determine which Core Courses will be used for Academic Continuation Evaluation. Please visit the calendar archives page and consult the School of Architecture for guidance.

Prohibited Courses

Students in the B.A.S. programs may not count any 0000-level courses for credit toward their degree. Such students may, however, be required to take one or more of these courses to replace missing program prerequisites in which case the courses will be set aside as “no credit for degree” (NCD).

Program Requirements

Design

B.A.S. Honours (20.0 credits)

A. Credits Included in the Major CGPA (16.5 credits)

1. 8.0 credits in core courses: 8.0

ARCH 1111 [1.0]	Studio 1A: Land
ARCH 1112 [1.0]	Studio 1B: Fundamentals of Design
ARCH 2111 [1.0]	Studio 2A: Fundamentals of Urbanism
ARCH 2172 [1.0]	Studio 2B: Local (Design)
ARCH 3111 [1.0]	Studio 3A: Adaptive Reuse
ARCH 3172 [1.0]	Studio 3B: Global (Design)
ARCH 4111 [1.0]	Studio 4A: Integrated
ARCH 4172 [1.0]	Studio 4B: Option (Design)

2. 8.5 credits in additional major requirements: 8.5

ARCH 1221 [0.5]	Material Histories of Architecture
ARCH 1222 [0.5]	Design, Climate, Environment
ARCH 1331 [0.5]	Introduction to Architecture
ARCH 1441 [0.5]	Drawing and Media
ARCH 1442 [0.5]	Digital Drawing and Modelling
ARCH 2221 [0.5]	Ecological & Regulatory Systems
ARCH 2222 [0.5]	Structures
ARCH 2331 [0.5]	Modernism and Global Urbanism
ARCH 2332 [0.5]	Architectures in Canada
ARCH 3221 [0.5]	Assemblies
ARCH 3331 [0.5]	Architectural Conservation Philosophy and Ethics
ARCH 3441 [0.5]	Digital Computation and Simulation
ARCH 4221 [0.5]	Environmental Systems

ARCH 4332 [0.5]	Contemporary Theories in Architecture
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ARCH 4771 [0.5]	Architectural Discourse and Methods
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ARCH 4772 [0.5]	The Cost of Building
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ARCH 4773 [0.5]	Designed Landscapes
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B. Credits Not Included in the Major CGPA (3.5 credits)

3. 1.5 credits in: 1.5

ARTH 1201 [0.5]	History and Theory of Architecture: 1500 to Present
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DBST 2001 [0.5]	Introduction to Disability Studies
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ARCH 4777 [0.5]	Land Ethics and Identities
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4. 0.5 credits from: 0.5

HRSJ 2001 [0.5]	Human Rights: Theories and Foundations
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HRSJ 2202 [0.5]	Power Relations and Human Rights
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5. 1.5 credits in free electives 1.5

Total Credits 20.0

Urbanism

B.A.S. Honours (20.0 credits)

A. Credits Included in the Major CGPA (16.5 credits)

1. 8.0 credits in the Urbanism Core: 8.0

ARCH 1111 [1.0]	Studio 1A: Land
ARCH 1112 [1.0]	Studio 1B: Fundamentals of Design
ARCH 2111 [1.0]	Studio 2A: Fundamentals of Urbanism
ARCH 2192 [1.0]	Studio 2B: Local (Urbanism)
ARCH 3111 [1.0]	Studio 3A: Adaptive Reuse
ARCH 3192 [1.0]	Studio 3B: Global (Urbanism)
ARCH 4111 [1.0]	Studio 4A: Integrated
ARCH 4192 [1.0]	Studio 4B: Option (Urbanism)

2. 8.5 credits in additional major requirements: 8.5

ARCH 1221 [0.5]	Material Histories of Architecture
ARCH 1222 [0.5]	Design, Climate, Environment
ARCH 1331 [0.5]	Introduction to Architecture
ARCH 1441 [0.5]	Drawing and Media
ARCH 1442 [0.5]	Digital Drawing and Modelling
ARCH 2221 [0.5]	Ecological & Regulatory Systems
ARCH 2222 [0.5]	Structures
ARCH 2331 [0.5]	Modernism and Global Urbanism
ARCH 2332 [0.5]	Architectures in Canada
ARCH 3221 [0.5]	Assemblies
ARCH 3331 [0.5]	Architectural Conservation Philosophy and Ethics

ARCH 3441 [0.5]	Digital Computation and Simulation
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ARCH 4221 [0.5]	Environmental Systems
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ARCH 4332 [0.5]	Contemporary Theories in Architecture
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ARCH 4991 [0.5]	History of Modern Housing
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ARCH 4992 [0.5]	Theories of Urbanism
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ARCH 4993 [0.5]	Topics in Urbanism
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B. Credits Not Included in the Major CGPA (3.5 credits)

3. 2.0 credits in: 2.0

ARTH 1201 [0.5]	History and Theory of Architecture: 1500 to Present
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GEOG 2023 [0.5]	Cities, Inequality and Urban Change
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GEOG 4323 [0.5]	Urban and Regional Planning	
GEOM 1004 [0.5]	Maps, Satellites and the Geospatial Revolution	
5. 1.5 credit in free electives		1.5
Total Credits		20.0

Conservation and Sustainability

B.A.S. Honours (20.0 credits)

A. Credits Included in the Major CGPA (16.5 credits)

1. 8.0 credits in core:		8.0
ARCH 1111 [1.0]	Studio 1A: Land	
ARCH 1112 [1.0]	Studio 1B: Fundamentals of Design	
ARCH 2111 [1.0]	Studio 2A: Fundamentals of Urbanism	
ARCH 2182 [1.0]	Studio 2B: Local (C&S)	
ARCH 3111 [1.0]	Studio 3A: Adaptive Reuse	
ARCH 3182 [1.0]	Studio 3B: Global (C&S)	
ARCH 4111 [1.0]	Studio 4A: Integrated	
ARCH 4182 [1.0]	Studio 4B: Option (C&S)	
2. 8.5 credits in additional major requirements:		8.5
ARCH 1221 [0.5]	Material Histories of Architecture	
ARCH 1222 [0.5]	Design, Climate, Environment	
ARCH 1331 [0.5]	Introduction to Architecture	
ARCH 1441 [0.5]	Drawing and Media	
ARCH 1442 [0.5]	Digital Drawing and Modelling	
ARCH 2221 [0.5]	Ecological & Regulatory Systems	
ARCH 2222 [0.5]	Structures	
ARCH 2331 [0.5]	Modernism and Global Urbanism	
ARCH 2332 [0.5]	Architectures in Canada	
ARCH 3221 [0.5]	Assemblies	
ARCH 3331 [0.5]	Architectural Conservation Philosophy and Ethics	
ARCH 3441 [0.5]	Digital Computation and Simulation	
ARCH 4221 [0.5]	Environmental Systems	
ARCH 4332 [0.5]	Contemporary Theories in Architecture	
ARCH 4881 [0.5]	Advanced Building Assessment	
ARCH 4882 [0.5]	Topics in Conservation	
ARCH 4883 [0.5]	Evaluation of Existing Properties	

B. Credits Not Included in the Major CGPA (3.50 credits)

3. 2.0 credits in:		2.0
ARTH 1201 [0.5]	History and Theory of Architecture: 1500 to Present	
CDNS 2400 [0.5]	Heritage Places and Practices in Canada	
ARCH 3881 [0.5]	Historic Site Recording and Assessment	
ARCH 4885 [0.5]	Building Pathology and Rehabilitation	
4. 1.5 credits in free electives		1.5
Total Credits		20.0

Regulations (B.A.S.)

In addition to the specific program requirements, students must satisfy the academic regulations of the university, and the faculty regulations for the degree, below. Students

should consult the School when planning their program and selecting courses.

Residency Requirement

B.A.S. Hons.

- Conservation and Sustainability
- Design
- Urbanism

To be eligible to graduate, students in these programs must present a minimum of 5.0 residency credits in their degree program.

For more information, consult Section 2.2.2/3.4.1 Minimum Number of Residency Credits (Residency and Advanced Credits) in the *Academic Regulations of the University* section of this Calendar.

Time Limit

The Bachelor of Architectural Studies degree must be completed within seven calendar years of initial registration. Students who do not complete their program within this limit will be withdrawn from the B.A.S. degree.

Retention of Work

Keeping a good portfolio is a most important part of architectural education. A portfolio represents a record of the student's progress and design experience over the years, and is an indispensable requirement for any future job application. A portfolio is started in first year and continues to expand until graduation. The School, therefore, requires that each student produce reproductions (normally 8 1/2 x 11 inch reproductions, colour or black and white, slides, and/or digital format CD) of his or her work at the end of each term. One copy of the work should be put in the student's portfolio and the other turned in to the instructor for retention in the School's archives. (This facilitates retrospective exhibitions of work, accreditation, publications and any future references for pedagogic purposes.) Original work is the property of the students, but the School retains the right to keep work of merit for up to two years after the date of submission. The School will make every effort to preserve the work in good condition, and will give authorship credit and take care of its proper use.

Academic Continuation Evaluation for Bachelor of Architectural Studies (Honours)

Students in the B.A.S. (Honours) follow the continuation requirements for Honours programs, as described in Section 3.2.6 of the *Academic Regulations of the University*, with the following addition:

- Students with 15.5 or more program credits completed, but who have a Major CGPA less than 6.00, will be required to leave the B.A.S. program with the decision *Required to Withdraw for Two Terms* (WT).

Co-operative Education

For more information about how to apply for the Co-op program and how the Co-op program works please visit the Co-op website.

All students participating in the Co-op program are governed by the Undergraduate Co-operative Education Policy.

Undergraduate Co-operative Education Policy

Admission Requirements

Students can apply to Co-op in one of two ways: directly from high school, or after beginning a degree program at Carleton.

If a student applies to a degree program with a Co-op option from high school, their university grades will be reviewed two terms to one year prior to their first work term to ensure they meet the academic requirements after their first or second year of study. The time at which the evaluation takes place depends on the program of study. Students will automatically receive an admission decision via their Carleton email account.

Students who did not request Co-op at the time they applied to Carleton can request Co-op after they begin their university studies. To view application instructions and deadlines, please visit carleton.ca/co-op.

To be admitted to Co-op, a student must successfully complete 5.0 or more credits that count towards their degree, meet the minimum CGPA requirement(s) for the student's Co-op option, and fulfil any specified course prerequisites. To see the unique admission and continuation requirements for each Co-op option, please refer to the specific degree programs listed in the Undergraduate Calendar.

Participation Requirements

Co-op Participation Agreement

All students must adhere to the policies found within the Co-op Participation Agreement.

COOP 1000

Once a student has been admitted to the Co-op Program, they will be given access to register in COOP 1000. This zero-credit online course must be completed at least two terms prior to the student's first work term.

Communication with the Co-op Office

Students must maintain contact with the Co-op Office during their job search and while on a work term. All email communication will be conducted via the students' Carleton email account.

Employment

Although every effort is made to ensure a sufficient number of job postings for all Co-op students, no guarantee of employment can be made. The Co-op job search process is competitive, and success is dependent upon factors such as current market conditions, academic performance, skills, motivation, and level of commitment to the job search. It is the student's responsibility to apply for positions via the Co-op job board in addition to actively conducting a self-directed job search. Students who do not obtain a co-op work term are expected to continue with their academic studies. It should be noted that hiring priority for positions within the Federal Government of Canada is given to Canadian citizens.

Registration

- Students must be registered as full-time during all fall and winter study terms beginning the term in which they enroll in COOP 1000.
- Students will be registered in a Co-op Work Term course while at work. This course does not carry academic course credit, but is noted on academic transcripts.
- Students may register in a 0.5 credit during a work term, provided the course is offered during the evening or is offered asynchronously online.
- Students must have at least one term of full-time studies left to complete following their final co-op work term. Students cannot end their degree on a work term.

Work Term Assessment and Evaluation

Work Term Evaluation

Employers are responsible for submitting to Carleton University final performance evaluations for their Co-op students at the end of their work terms.

Work Term Assessment

In order to successfully complete the co-op work term, students must receive a Satisfactory (SAT) grade on their Co-op Work Term Report, which they must submit at the completion of each four-month work term.

Graduation with the Co-op Designation

In order to graduate with the Co-op Designation, students must satisfy all requirements of the degree program in addition to the successful completion of three or four work terms (the number is dependent upon the student's academic program). Students found in violation of the Co-op Participation Agreement may have the Co-op Designation withheld.

Note: Participation in the co-op option will add up to one additional year for a student to complete their degree program.

Voluntary Withdrawal from the Co-op Option

Students who are currently on a co-op work term or who have already committed to a co-op work term either verbally or in writing may not leave the position and/or withdraw from the co-op option until they have completed the work term and all related requirements.

Involuntary or Required Withdrawal from the Co-op Option

Students may be removed from the Co-op Program for any of the following reasons:

1. Failure to achieve a grade of SAT in COOP 1000;
2. Failure to attend all interviews for positions to which the student has applied;
3. Declining more than one job offer during the job search;
4. Reneging on a co-op position that the student has accepted either verbally or in writing;
5. Continuing a job search after accepting a co-op position;

6. Dismissal from a work term by the co-op employer;
7. Leaving a work term without approval from the Co-op Management Team;
8. Receipt of an unsatisfactory work term evaluation;
9. Receiving a grade of UNS on the work term report.

International Students

All international students are required to possess a Co-op Work Permit issued by Immigration, Refugees and Citizenship Canada before they can begin working. The Co-operative Education Office will provide students with a letter of support to accompany their Co-op Work Permit application. Students are advised to discuss the application process and application requirements with the International Student Services Office.

Co-op Fees

All participating Co-op students are required to pay Co-op fees. For full details, please see the Co-op website.

Bachelor of Architectural Studies: Co-op Admission and Continuation Requirements

- Maintain full-time status in each study term;
- Be eligible to work in Canada (for off-campus work);
- Have successfully completed COOP 1000 .

In addition to the following:

1. Registered as a full-time student in the B.A.S. program;
2. Obtained third-year standing;
3. Obtained an Overall CGPA of at least 8.0. This CGPA must be maintained throughout the duration of the degree.

B.A.S. students must successfully complete three (3) work terms to obtain the Co-op Designation.

Work Term Report Course: ARCH 3999 [0.0]
Work/Study Pattern:

Year 1		Year 2		Year 3		Year 4		Year 5	
Term	Pattern	Term	Pattern	Term	Pattern	Term	Pattern	Term	Pattern
Fall	S	Fall	S	Fall	S	Fall	W	Fall	S
Winter	S	Winter	S	Winter	S	Winter	W	Winter	S
Summer		Summer		Summer	W	Summer	W		

Legend

S: Study

W: Work

Admissions Information

Admission Requirements are for the 2026-27 year only, and are based on the Ontario High School System. Holding the minimum admission requirements only establishes eligibility for consideration. The cut-off averages for admission may be considerably higher than the minimum. See also the **General Admission and Procedures** section of this Calendar. An overall average of at least 70% is normally required to be considered for admission. Some programs may also require specific course prerequisites and prerequisite averages and/or supplementary admission portfolios. Higher averages are required for admission to programs for which the demand for places by qualified applicants exceeds the number of places available. The overall average required

for admission is determined each year on a program by program basis. Consult admissions.carleton.ca for further details.

Note: Courses listed as *recommended* are not mandatory for admission. Students who do not follow the recommendations will not be disadvantaged in the admission process.

Admissions Information

Admission requirements are based on the Ontario High School System. Prospective students can view the admission requirements through the Admissions website at admissions.carleton.ca. The overall average required for admission is determined each year on a program-by-program basis. Holding the minimum admission requirements only establishes eligibility for consideration; higher averages are required for admission to programs for which the demand for places by qualified applicants exceeds the number of places available. All programs have limited enrolment and admission is not guaranteed. Some programs may also require specific course prerequisites and prerequisite averages and/or supplementary admission portfolios. Consult admissions.carleton.ca for further details.

Note: If a course is listed as recommended, it is not mandatory for admission. Students who do not follow the recommendations will not be disadvantaged in the admission process.

Degree

- Bachelor of Architectural Studies Honours (B.A.S.)

Admission Requirements

First Year

The Ontario Secondary School Diploma (OSSD) or equivalent including a minimum of six 4U or M courses. The six 4U or M courses must include English, Physics, and Advanced Functions. Calculus and Vectors is strongly recommended.

Note: a portfolio is required. Detailed information about the portfolio requirements can be found on the Undergraduate Admissions website at admissions.carleton.ca.

Advanced Standing

Applications for admission to the second or subsequent years will be assessed on their merits. Applicants must normally be *Eligible to Continue* in their year level, in addition to meeting the CGPA thresholds described in Section 3.1.9 of the *Academic Regulations of the University*.

Applicants will also be required to complete a portfolio which will assist in the evaluation of their suitability for the program. Detailed information about the portfolio requirements can be found at admissions.carleton.ca.

Students will not receive credit for courses graded below C-.

Co-op Option

Direct Admission to the First Year of the Co-op Option
 Applicants must:

1. meet the required overall admission cut-off average and prerequisite course average. These averages may be higher than the stated minimum requirements;
2. be registered as a full-time student in the B.A.S. program;
3. be eligible to work in Canada (for off-campus work placements).

Meeting the above requirements only establishes eligibility for admission to the program. The prevailing job market may limit enrolment in the co-op option.

Note: continuation requirements for students previously admitted to the co-op option and admission requirements for the co-op option after beginning the program are described in the Co-operative Education Regulations section of this Calendar.

Architecture (ARCS) Courses

ARCS 3105 [1.5 credit]

Studio 4

Supported by the core curriculum, focuses on a medium-scale building within a regional context. May include a small design-build. Projects further analog and digital methods. May introduce concepts like adaptive re-use while furthering the understanding of structure and building systems in a complex building. (Core Course).

Includes: Experiential Learning Activity

Precludes additional credit for ARCH 3111.

Prerequisite(s): ARCS 2105 and ARCS 2106.

Twelve hours studio, plus one hour lecture per week.

ARCS 3107 [1.0 credit]

Studio 5

The Directed Studies Abroad (DSA) studio considers large-scale, mixed-use buildings in an international context. Design projects advance analog and digital methods to explore broader cultural and social conditions within a complex site often in conjunction with a site visit abroad. (Core Course).

Includes: Experiential Learning Activity

Precludes additional credit for ARCH 3172, ARCH 3182, ARCH 3192, ARCS 3106 (no longer offered).

Prerequisite(s): ARCS 2105 and ARCS 2106.

Eight hours studio, plus one hour lecture per week.

ARCS 3301 [1.0 credit]

Conservation Studio 2

Historical building projects exploring architecture as a form of cultural expression. Consideration of site, program and materials. Introduction of conservation, sustainability and adaptive re-use principles, development standards, architectural codes, using case studies in Ottawa and elsewhere. Physical, digital drawings and models to explore designs. (Core).

Includes: Experiential Learning Activity

Precludes additional credit for ARCH 3111, ARCC 3301 (no longer offered).

Prerequisite(s): ARCS 2302 and third-year standing in B.A.S. Conservation and Sustainability major or permission of the School.

Studio eight hours per week.

ARCS 3302 [1.0 credit]

Conservation Studio 3

The role of architecture in culture, stressing site and program with respect to their historic, social and ecological implications. Synthesis of issues, methods and techniques of the conservation and sustainability curriculum. (Core Course).

Includes: Experiential Learning Activity

Precludes additional credit for ARCC 3302 (no longer offered), ARCH 3172, ARCH 3182, ARCH 3192.

Prerequisite(s): ARCS 3301 and third-year standing in B.A.S. Conservation and Sustainability major or permission of the School.

Studio eight hours per week.

ARCS 3304 [1.0 credit]

Urbanism Studio 3: Urbanism on the Periphery

Urbanization, sprawl, growth models, land consumption, containment strategies (smart growth, greenbelts, growth boundaries), edge cities, the Just City, Ecological Urbanism, and informal suburbanization in developed and developing countries. Through design, students explore the impact of practices, pressures, processes and cultural preferences on the expanding city.

Includes: Experiential Learning Activity

Precludes additional credit for ARCH 3111, ARCU 3304 (no longer offered).

Prerequisite(s): ARCS 2303 and ARCS 2304 and third-year standing in B.A.S. Urbanism major or permission of the School.

Eight hours studio, plus one hour lecture per week.

ARCS 3306 [1.0 credit]**Urbanism Studio 5: Global Perspectives**

Urbanization as a global phenomenon. Study of various forms of urbanization and de-urbanization in relation to economic, political and cultural forces. Through design, students explore the (trans)formation of settlements and communities outside of the North American context.

Includes: Experiential Learning Activity

Precludes additional credit for ARCH 3172, ARCH 3182, ARCH 3192, ARCS 4304 (no longer offered).

Prerequisite(s): ARCS 2303 and ARCS 2304 and third-year standing in B.A.S. Urbanism major or permission of the School.

Eight hours studio, plus one hour lecture per week.

ARCS 4105 [1.5 credit]**Comprehensive Studio**

Focussing on multi-unit housing, students from BAS majors collaborate to develop strategies for redevelopment of large urban sites. Engages urban design, site planning, programming, adaptive reuse, and community consultation. Students produce detailed designs for buildings, emphasizing building systems and envelope design. (Core Course).

Includes: Experiential Learning Activity

Precludes additional credit for ARCH 4111.

Prerequisite(s): ARCS 3105 and ARCS 3107, or ARCS 3303 and ARCS 3304.

Twelve hours studio, plus one hour lecture per week.

ARCS 4107 [1.0 credit]**Option Studio**

Offers a range of topics for exploration. Students use analog and digital methods and techniques to culminate the undergraduate studio sequence while offering focused research-led investigation into key social, political, spatial issues. (Core Course).

Includes: Experiential Learning Activity

Precludes additional credit for ARCH 4172, ARCH 4182, ARCH 4192, ARCS 4106 (no longer offered).

Prerequisite(s): ARCS 3105 and ARCS 3107.

Eight hours studio, plus one hour lecture per week.

Architecture (ARCC) Courses**ARCC 2203 [0.5 credit]****Architectural Technology 3**

Wood frame, post and beam, steel and concrete systems and construction techniques. Structural systems and building envelope principles and practise are explored in conjunction with mechanical and electrical systems in smaller buildings. Emphasis on precedent, tradition and methodology of architectural detailing for construction.

Includes: Experiential Learning Activity

Precludes additional credit for ARCH 3221.

Prerequisite(s): ARCC 2202 and third-year standing for B.A.S. students and third-year standing for students in B.Eng. Architectural Conservation and Sustainability.

Lectures three hours a week.

ARCC 3202 [0.5 credit]**Architectural Technology 4**

Medium scale steel, concrete, and wood frame buildings as case studies to explore approaches to building science principles, building envelope design, advanced construction methods and materials, acoustics and sound control, and fire protection. Focus on sustainable design strategies and environment impact. (Core course).

Precludes additional credit for ARCH 4221.

Prerequisite(s): ARCC 2203 and third-year standing for B.A.S. students or ARCC 2203 and third-year standing for students in B.Eng. Architectural Conservation.

Lectures three hours a week.

ARCC 4200 [0.5 credit]**Structural Morphology**

Interdisciplinary study of structural and developmental morphology focusing on dynamic generative design processes, integrative systems, spatial modulations and fundamental generative principles of spatial form and structure as it relates to architecture. (Workshop).

Includes: Experiential Learning Activity

Lectures, seminar, workshop or field work six hours a week.

ARCC 4207 [0.5 credit]**Advanced Building Assessment**

In-depth study of the conventions, methods, and tools used in the assessment of buildings and their sites including traditional field survey, photogrammetry, laser scanning technologies, and hybrid representations.

Includes: Experiential Learning Activity

Precludes additional credit for ARCC 4900 (no longer offered).

Prerequisite(s): enrolment in the BAS Conservation and Sustainability program and fourth-year standing.

Laboratories, lectures, field trips, six hours a week.

ARCC 4500 [0.5 credit]**Design Economics**

Principles of building economics. Determinants and prediction of building costs. Uncertainty and investment economics. Creative cost control for buildings during schematic design, design development, construction document preparation and construction. Economic evaluation during all phases of design process; emphasis on sustainable strategies.

Precludes additional credit for ARCC 3500, ARCH 4772.

Prerequisite(s): fourth-year standing in the B.A.S. program or permission of the School.

Three hours a week.

ARCC 4801 [0.5 credit]**Architectural Technology**

A specific aspect of architecture in the area of architectural technology. Topics vary from year to year. (Elective Course).

Prerequisite(s): permission of the School.

Architecture (ARCN) Courses**ARCN 4100 [0.5 credit]****Historic Site Recording and Assessment**

Methods of heritage building documentation including hand recording, photography, rectified photography, total station, gps, photogrammetry, and laser scanning. Non-destructive testing techniques; environmental assessment tools for determining air quality and energy efficiency.

Multidisciplinary teams for all project work.

Includes: Experiential Learning Activity

Also listed as ACSE 3207, CIVE 3207.

Precludes additional credit for ARCN 3100 (no longer offered), ARCH 3881.

Prerequisite(s): second-year standing in B.A.S.

Conservation and Sustainability.

Lectures three hours a week, lab or field work two hours a week.

ARCN 4200 [0.5 credit]**Building Pathology and Rehabilitation**

Deterioration mechanisms for concrete, timber, steel and masonry structures. Identification of design deficiencies; criteria for selection and design of rehabilitation systems. Design techniques to reduce deterioration in new construction and historical structures.

Includes: Experiential Learning Activity

Also listed as ACSE 4601, ARCH 4885, CIVE 4601.

Prerequisite(s): ARCN 4100 and third-year standing in B.A.S.

Conservation and Sustainability.

Lectures three hours a week, lab/field work two hours a week.

Architecture (ARCH) Courses**ARCH 1007 [0.5 credit]****Introduction to Drawing: Seeing Through the Hand**

Fundamental concepts of line and line weight, light and shadow, perspective, contrast and composition. Exercises will include some mixed media and will introduce students to drawing as a way of translating ideas into images.

Includes: Experiential Learning Activity

Precludes additional credit for ARCN 1005 (no longer offered).

One hour lecture and two hours drawing/discussion.

ARCH 1111 [1.0 credit]**Studio 1A: Land**

Studio course involving land-based workshops and fieldwork to introduce orientation, siting, topography, land work, material tectonic and building foundations. Students learn drawing conventions, architectural drafting, and physical modeling, applying basic spatial norm and sequencing through the design of a small-scale building in a non-urban context.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 1005.

Prerequisite(s): Registration in the Bachelor of Architectural Studies (BAS) program.

Studio 10 hours a week.

ARCH 1112 [1.0 credit]**Studio 1B: Fundamentals of Design**

Studio course considering siting, orientation, and building design. With hybrid drawings and models, students move from spatial abstraction to inhabitation. Designing a small domestic program on an urban site, they learn basic programmatic organization, accessibility, spatial hierarchy, material selection, structural systems, and envelope design.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 1105.

Prerequisite(s): Minimum grade of C- or above in ARCH 1111.

Studio 10 hours a week.

ARCH 1221 [0.5 credit]**Material Histories of Architecture**

Historical survey of architecture and conservation through material practices, transformation, and innovation around the world. Study of thousand-year old methods, monuments, and heritage sites, following materials including clay, metals, wood, or concrete, and studying their relations to landscapes, built forms, cultures, and climate.

Precludes additional credit for ARCC 1202 (no longer offered), ARCC 3502 (no longer offered).

Prerequisite(s): Registration in the Bachelor of Architectural Studies (BAS) program.

Three hour lecture

ARCH 1222 [0.5 credit]**Design, Climate, Environment**

Environmental histories following the migration of peoples, practices, and resources across territories and times to introduce buildings' climate impacts and environmental design. Consideration of construction principles, energy management, material selection, building siting, and environmental control design along larger consideration of climate and environmental justice. Precludes additional credit for ARCC 2100 (no longer offered).

Prerequisite(s): Registration in the Bachelor of Architectural Studies (BAS) program.

Three hour lecture

ARCH 1331 [0.5 credit]**Introduction to Architecture**

Introduction of architecture from the perspective of land and climate, examining social, cultural, and environmental relationships between peoples, places, and practices. Consideration of Indigenous land rights, topographical conditions and land formation, stratigraphy and soil composition, landmarks and foundations, placement and displacement, lived-experience and land-based practices. Precludes additional credit for ARCH 1000 (no longer offered).

Three hour lecture

ARCH 1441 [0.5 credit]**Drawing and Media**

Introduction to various representational media, including orthographic drawings and alternative multimedia techniques. Historical, theoretical, and practical explorations of visual communication, moving between analogue, digital, and graphic image making processes to document, develop, and communicate sites or design projects. Includes assignments conducted in parallel with studio.

Includes: Experiential Learning Activity

Precludes additional credit for ARCN 2106 (no longer offered).

Prerequisite(s): Registration in the Bachelor of Architectural Studies (BAS) program.

Three hour lecture + three hour lab

ARCH 1442 [0.5 credit]**Digital Drawing and Modelling**

Introduction to the logics of computer software for digital drawing, modeling, and visual coding. Extensive practical work using appropriate applications. Includes assignments conducted in parallel with studio, typically incorporating case study analyses.

Includes: Experiential Learning Activity

Precludes additional credit for ARCN 2105 (no longer offered).

Prerequisite(s): Registration in the Bachelor of Architectural Studies (BAS) program.

Three hour lecture + three hour lab

ARCH 2006 [0.5 credit]**Theory and History of Design**

The theoretical and historical background of industrial design and design; disciplinary foundations and interdisciplinary connections; methodological aspects and economic and social contexts; contemporary scenarios in design; technological innovation and manufacturing processes. (Elective course).

Also listed as IDES 1000.

Lectures three hours a week.

ARCH 2101 [0.5 credit]**Industrial Design Analysis**

Principles of comparative product design analysis covering marketing and sales, manufacturing techniques and materials, ambiance and qualities of the object/context relationship, and design analysis from the perspective of the designer, the end-user and the environment. (Elective course).

Includes: Experiential Learning Activity

Also listed as IDES 1001.

Prerequisite(s): ARCH 2006 or IDES 1000.

Lectures three hours a week.

ARCH 2111 [1.0 credit]**Studio 2A: Fundamentals of Urbanism**

Urbanism studio introducing, documenting, and analyzing forces producing urban environments. Students design a medium-scale public infrastructure project in a local urban context, foregrounding site analysis and urban design skills. Consideration given to accessibility, public realm, and to broad impact of infrastructural, environmental, and ecological systems.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 2105 (no longer offered), ARCS 2303 (no longer offered).

Prerequisite(s): A minimum grade of C- or above in ARCH 1111 and ARCH 1112.

Studio 10 hours a week.

ARCH 2172 [1.0 credit]
Studio 2B: Local (Design)

Design studio working with community with appropriate ethics training on a small to medium building project. Students use analog and digital methods to advance consideration of site, program, and materials as the means for shaping the built environment, understanding the settings, and their communities.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 2106 (no longer offered), ARCS 2302 (no longer offered), ARCS 2304 (no longer offered).

Prerequisite(s): Minimum grade of C- or above in ARCH 2111 and Registration in the Bachelor of Architectural Studies (BAS) Design stream.

Studio 10 hours a week.

ARCH 2182 [1.0 credit]
Studio 2B: Local (C&S)

Conservation & Sustainability studio working with community with appropriate ethics training on a small to medium building projects. Students test and study conservation methodologies through design exercises and historical research on existing architectures, cities, and landscapes, with emphasis on understanding the settings and its communities.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 2106 (no longer offered), ARCS 2302 (no longer offered), ARCS 2304 (no longer offered).

Prerequisite(s): Minimum grade of C- or above in ARCH 2111 and Registration in the Bachelor of Architectural Studies (BAS) Conservation and Sustainability stream.

Studio 10 hours a week

ARCH 2192 [1.0 credit]
Studio 2B: Local (Urbanism)

Urbanism studio working with community with appropriate ethics training. Students design small scale projects exploring ramifications of practices, policies, and cultural preferences on urban cores. Consideration of intensification, revitalization, gentrification, brownfield redevelopment, development standards, form-based codes, and the larger impact of migration on urban density.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 2106 (no longer offered), ARCS 2302 (no longer offered), ARCS 2304 (no longer offered).

Prerequisite(s): Minimum grade of C- or above in ARCH 2111 and Registration in the Bachelor of Architectural Studies (BAS) Urbanism stream.

Studio 10 hours a week

ARCH 2221 [0.5 credit]
Ecological & Regulatory Systems

Lecture-based course considering buildings' ecological and climate change impacts. Course covers how regulations, material choices, siting, and assembly methods of small-scale buildings impact energy consumption, carbon footprints, and GHG emissions. Regulatory content includes applicable codes, regulations, best practices, universal design standards and life-safety systems principles.

Includes: Experiential Learning Activity

Precludes additional credit for ARCC 2202 (no longer offered), ARCC 5096 (no longer offered).

Prerequisite(s): ARCH 1221 and ARCH 1222.

Also offered at the graduate level, with different requirements, as ARCH 5221, for which additional credit is precluded.

Three hour lecture + three hour lab

ARCH 2222 [0.5 credit]
Structures

Fundamental structural principles and their building design applications. Concepts of equilibrium and mechanics of materials, including stress and strain. Structural calculations and qualitative understanding of static and dynamic loads, including gravitational and lateral forces. Includes consideration of wood, masonry, concrete, and steel structural systems.

Precludes additional credit for CIVE 2005, ARCC 5097.

Prerequisite(s): ARCH 2221.

Also offered at the graduate level, with different requirements, as ARCH 5222., for which additional credit is precluded.

Three hour lecture

ARCH 2331 [0.5 credit]
Modernism and Global Urbanism

Thematic survey of cities, modernist ideas, projects, and movements, considering their theoretical, historical, and practical expressions in urban morphology and housing typologies, and their relation to larger societal and environmental questions, locally and globally. Acquisition of critical reading, writing, and representation skills through case-study analysis.

Precludes additional credit for ARCH 2300 (no longer offered), ARCU 3100.

Prerequisite(s): ARTH 1201.

Also offered at the graduate level, with different requirements, as ARCH 5331., for which additional credit is precluded.

Three hour lecture

ARCH 2332 [0.5 credit]**Architectures in Canada**

Architectures in Canada, including Indigenous settlements, practices, and relationships to the land to this day. Survey of selected buildings to consider relational, symbolic, stylistic, and technological developments. Critical analysis of styles, methods, materials, and building typologies from social, cultural, economic, and constructional perspectives.

Precludes additional credit for ARCH 4002.

Prerequisite(s): ARCH 2331.

Three hour lecture

ARCH 3111 [1.0 credit]**Studio 3A: Adaptive Reuse**

Adaptive architecture studio critically considering buildings' adaptability in diverse communities' contexts and in relation to environmental responsibilities.

Standards, principles, basic regulatory systems, and codes of conservation introduced through case-studies and the designing a medium-scale project, with focus on program analysis, detailing and material assembly.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 3105, ARCS 3301, ARCS 3304.

Prerequisite(s): Minimum grade of C- or above in ARCH 2111.

Studio 10 hours a week.

ARCH 3172 [1.0 credit]**Studio 3B: Global (Design)**

The global partnership Design studio (DSA) considers the role of design in communities, stressing site and program with respect to their historic, social, and ecological implications in a mid-scale cultural building. Synthesis and expansion of issues, methods, and techniques of the design curriculum.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 3302, ARCS 3107, ARCS 3306.

Prerequisite(s): Minimum of a C- or above in ARCH 3111.

Studio 10 hours a week.

ARCH 3182 [1.0 credit]**Studio 3B: Global (C&S)**

The global partnership Conservation & Sustainability studio (DSA) considers the role of design in communities, stressing site and program with respect to their historic, social, and ecological implications in a mid-scale cultural building. Synthesis of issues, methods, and techniques of the conservation and sustainability curriculum.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 3302, ARCS 3107, ARCS 3306.

Prerequisite(s): Minimum grade of C- or higher in ARCH 3111 and Registration in the Conservation and Sustainability stream of the Bachelor of Architectural Studies.

Studio 10 hours a week.

ARCH 3192 [1.0 credit]**Studio 3B: Global (Urbanism)**

The global partnership Urbanism studio (DSA) considers the role various forms of urbanization and de-urbanization in relation to economic, political, and cultural forces.

Students design a mid-scale public building, synthesizing issues, methods, and techniques of the urbanism curriculum, learning about urbanization as a global phenomenon.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 3302, ARCS 3107, ARCS 3306.

Prerequisite(s): Minimum grade of C- or above in ARCH 3111.

Studio 10 hours a week

ARCH 3221 [0.5 credit]**Assemblies**

Wood, steel, concrete, and alternate construction materials, and systems. Building envelope principles and practices explored in conjunction with mechanical and electrical systems in buildings. Emphasis on precedents, traditions, and methodology of architectural detailing for construction and adaptation. Introduction of environmental impact assessment and life-cycle analysis.

Precludes additional credit for ARCC 2203, ARCH 5223.

Prerequisite(s): ARCH 2222.

Three hour lecture

ARCH 3331 [0.5 credit]**Architectural Conservation Philosophy and Ethics**

Analysis of philosophical theories and related approaches to the material transformation of buildings. Micro-histories in architectural conservation theory and practice; overview of historical and contemporary concepts in architectural conservation. Preservation, restoration, rehabilitation, reconstruction, adaptive re-use, conservation anamnesis, diagnosis.

Precludes additional credit for ARCH 4200.

Prerequisite(s): Third year standing in the Bachelor of Architectural Studies or the Bachelor of Engineering in Architectural Conservation and Sustainability Engineering.
Three hour lecture

ARCH 3441 [0.5 credit]**Digital Computation and Simulation**

Intermediate computer drawing and modeling with a focus on visualization, simulation, computation, and coding. Assignments conducted in parallel with studio, includes an introduction to Building Information Modeling and building documentation strategies and technologies.

Includes: Experiential Learning Activity

Prerequisite(s): ARCH 1442.

Three hour lecture + three hour lab

ARCH 3601 [0.5 credit]**Architectural Discourse I**

Examines ideas relevant to contemporary architectural discourses and practices focused on the development of critical thinking and communication skills situated in emerging inquiries within a longer lineage of existing architectural theory. (Core Course).

Precludes additional credit for ARCH 4771.

Prerequisite(s): Third-year standing or permission of the School.

Lecture 3 hours per week

ARCH 3881 [0.5 credit]**Historic Site Recording and Assessment**

Methods of heritage building documentation including hand recording, photography, rectified photography, total station, gps, photogrammetry, and laser scanning. Non-destructive testing techniques; environmental assessment tools for determining air quality and energy efficiency. Multidisciplinary teams for all project work.

Includes: Experiential Learning Activity

Also listed as ACSE 3207, CIVE 3207.

Precludes additional credit for ARCN 4100.

Prerequisite(s): Second-year standing in B.A.S.

Conservation and Sustainability stream.

Lectures three hours a week, lab or field work two hours a week.

ARCH 3999 [0.0 credit]**Co-operative Work Term**

Co-operative Work Term for students in the Bachelor of Architectural Studies.

Includes: Experiential Learning Activity

ARCH 4002 [0.5 credit]**Canadian Architecture**

Canadian architecture from the seventeenth century to the present. Building styles, methods, construction techniques, and materials in the context of social and economic conditions of both indigenous and settlement approaches to the built environment.

Includes: Experiential Learning Activity

Also listed as ARTH 3002.

Precludes additional credit for ARCH 2332, ARCH 3002.

Prerequisite(s): ARCH 2300 or permission of the School.

Lectures, seminars three hours a week.

ARCH 4105 [0.5 credit]**Theories of Landscape Design**

Introduction to landscape architecture as the organization of outdoor space. Historical, cultural, economic and political factors as a basis for interpreting spatial organization in urban and rural areas of human settlement. Emphasis on the period from the fifteenth to the nineteenth century. (Theory/History Elective).

Precludes additional credit for ARCH 4773.

Prerequisite(s): second-year standing or above.

Lectures three hours a week.

ARCH 4111 [1.0 credit]**Studio 4A: Integrated**

Integrated studio working in parallel with technology course to support students as they design a sustainable mixed-use housing project. Consideration of site planning, programming, regulatory systems, materials, and structures in a comprehensive building design incorporating environmental and structural systems as well as detailed envelope design.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 4105, ARCS 4301 (no longer offered).

Prerequisite(s): Minimum of a C- or above in ARCH 3111.

Studio 10 hours a week

ARCH 4172 [1.0 credit]**Studio 4B: Option (Design)**

Options of Design topics for speculative exploration in preparation for graduate studies. Projects can range in scale, and locations. Topics can variously expand upon acquired technical, historical, theoretical, or technology skills through focused research-led investigations into key spatial, material, social, or environmental issues.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 4107, ARCS 4302 (no longer offered).

Prerequisite(s): Minimum of a C- or above in ARCH 4111.

Studio 10 hours a week

ARCH 4182 [1.0 credit]**Studio 4B: Option (C&S)**

Options of Conservation and Sustainability topics for speculative exploration in preparation for graduate studies. Projects can range in scale, and locations. Topics can variously expand upon acquired technical, historical, theoretical, or technology skills through focused research-led investigations into key spatial, material, social, or environmental issues.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 4107, ARCS 4302 (no longer offered).

Prerequisite(s): Minimum of a C- or above in ARCH 4111.

Studio 10 hours a week

ARCH 4192 [1.0 credit]**Studio 4B: Option (Urbanism)**

Options of Urbanism topics for speculative exploration in preparation for graduate studies. Projects can range in scale, and locations. Topics can variously expand upon acquired technical, historical, theoretical, or technology skills through focused research-led investigations into key spatial, material, social, or environmental issues.

Includes: Experiential Learning Activity

Precludes additional credit for ARCS 4107, ARCS 4302 (no longer offered).

Prerequisite(s): Minimum of a C- or above in ARCH 4111.

Studio 10 hours a week

ARCH 4200 [0.5 credit]**Architectural Conservation Philosophy and Ethics**

Analysis of philosophical theories and related approaches to the material transformation of buildings. Micro-histories in architectural conservation theory and practice; overview of historical and contemporary concepts in architectural conservation. Preservation, restoration, rehabilitation, reconstruction, adaptive re-use, conservation anamnesis, diagnosis.

Precludes additional credit for ARCH 3100 (no longer offered), ARCH 3331.

Prerequisite(s): Third-year standing in B.A.S.; OR third-year status in B.Eng. (Architectural Conservation and Sustainability).

Lectures three hours a week.

ARCH 4201 [0.5 credit]**History of Modern Housing**

Study of housing as a function of social organization, demographics, market demand and public policy. Topics include the evolution of housing form, the role of the state, and the participation of architects in the housing marketplace in the 19th and 20th century. (Theory/History Elective).

Precludes additional credit for ARCH 4991.

Prerequisite(s): third-year standing in the B.A.S. program or permission of the School.

Lectures three hours a week.

ARCH 4206 [0.5 credit]**Recycling Architecture in Canada and Abroad**

Concepts of mediating old and new architecture at the scale of the city through to the detail of the construction joint. Issues in sustainability and cultural identity illuminated by recycled architecture and adaptive reuse are explored through readings, drawings and case studies. (Theory/History Elective).

Prerequisite(s): third-year standing in the B.A.S. program or by permission of the instructor or fourth-year standing in the B.Eng. Architectural Conservation and Sustainability program.

Lectures three hours a week.

ARCH 4221 [0.5 credit]**Environmental Systems**

Lecture-based technology course reinforcing building science principles of environmental mediation through building envelope, structural systems, passive and active systems, material selection, MEP, daylight, and acoustic. Consideration of fire protection, life-safety, climate adaptation and mitigation through life-cycle analysis, energy, and performance assessment in integrated studio project.

Precludes additional credit for ARCC 3202.

Prerequisite(s): ARCH 3221.

Three hour lecture

ARCH 4301 [0.5 credit]**Post-War Architecture**

Theoretical, ideological and artistic debates that have influenced the development of world architecture since 1950. (Theory/History Elective).

Also listed as ARTH 4604.

Prerequisite(s): ARCH 2300 or ARTH 3609 or permission of the instructor.

Lecture or seminar three hours per week.

ARCH 4332 [0.5 credit]**Contemporary Theories in Architecture**

Survey of cultural theories from the beginning of colonialism to this day. Considerations of how technological, socio-political, material, and ecological transformations inform architectural discourse. Students acquire research skills, considering topics such as race, gender, disability, environmental justice, Indigenous worldviews, climate, decolonization, or artificial intelligence.

Precludes additional credit for ARCH 5020, ARCH 4601.

Prerequisite(s): Fourth year standing in the Bachelor of Architectural Studies program.

Also offered at the graduate level, with different requirements, as ARCH 5332., for which additional credit is precluded.

Three hour lecture

ARCH 4505 [0.5 credit]**Seminar in Theory and History**

History and theory of architecture. Topics will vary from year to year. Limited enrolment. (Elective Course).

Prerequisite(s): fourth-year standing in the B.A.S. or B.A. (Honours) Architecture/Art History programs, or permission of the School.

Lectures three hours a week.

ARCH 4601 [0.5 credit]**Architectural Discourse II**

Examines ideas and methods relevant to contemporary architectural discourse with a focus on cultural diversity and global perspectives. Architectural Discourse II builds on learned skills from previous work and acts as a preparatory course for research skills necessary at the graduate level. (Core Course).

Precludes additional credit for ARCH 4332.

Prerequisite(s): ARCH 3601 and fourth-year standing or permission of the School.

Lecture three hours per week.

ARCH 4771 [0.5 credit]**Architectural Discourse and Methods**

Survey of ideas and methods relevant to contemporary architectural discourse and practices, selected to represent a broad range of approaches and perspectives. Development of critical thinking and communication skills, and introduction to design research methods along with those across humanities, sciences, and social sciences.

Precludes additional credit for ARCH 3601.

Prerequisite(s): Third year standing in the Bachelor of Architectural Studies (BAS) program.

Three hour lecture

ARCH 4772 [0.5 credit]**The Cost of Building**

The course explores the social, environmental, and economic costs of building. Topics range from pro-forma exercises for individual buildings to explorations of supply chains, environmental impact of various construction materials and methods, and the social impact of development and displacement on vulnerable sites and communities.

Precludes additional credit for ARCC 4500.

Prerequisite(s): Fourth year standing in the Bachelor of Architectural Studies (BAS) program.

Three hour lecture

ARCH 4773 [0.5 credit]**Designed Landscapes**

Introduction to the complex nature of designed landscapes, their material, socio-political, and historical relationships, their tensions, implications, plural geographies, and cultures. Topics include cross-disciplinarity and links with other design fields, issues of climate and adaptation, urbanism and infrastructure, representation and visibility, time and place.

Precludes additional credit for ARCH 4105.

Prerequisite(s): Third year standing or above.

Three hour lecture

ARCH 4777 [0.5 credit]**Land Ethics and Identities**

Exploration of land ethics and identities in relation to social, culture, political, and economic forces, through an investigation of the built and unbuilt environments. Topics include the setting of human activities, indigenous and non-indigenous relations, indigenous knowledges, geopolitics, sovereignty, settler-colonialism, and shifting identity politics.

Three hour lecture

ARCH 4801 [0.5 credit]**Special Topics**

An aspect of architecture in the area of theory and history. Topics vary from year to year. (Theory/History Elective).

Prerequisite(s): ARCH 2300 or permission of the School.

Lectures three hours a week.

ARCH 4808 [0.5 credit]
Independent Study
(Elective Course).

ARCH 4881 [0.5 credit]

Advanced Building Assessment

In-depth study of the conventions, methods, and tools used in the assessment of buildings and their sites including traditional field survey, photogrammetry, laser scanning technologies, and hybrid representations. Includes: Experiential Learning Activity
Also offered at the graduate level, with different requirements, as ARCH 5404., for which additional credit is precluded.
Three hour lecture

ARCH 4882 [0.5 credit]

Topics in Conservation

Advanced seminar in conservation and sustainability. Topics may include histories and theories related to adaptive architecture, heritage considerations, and critical approaches to conservation of buildings, cities, and landscapes.
Three hour lecture

ARCH 4883 [0.5 credit]

Evaluation of Existing Properties

The cultural, political, economic, and legal factors that shape our definition of and approaches to existing architecture. Processes for and implications of heritage designation, cultural value, and costs associated with restoration and ongoing preservation of heritage and other existing properties. Includes: Experiential Learning Activity
Prerequisite(s): Fourth year standing in the Bachelor of Architectural Studies (BAS) Conservation & Sustainability stream.
Also offered at the graduate level, with different requirements, as ARCH 5402., for which additional credit is precluded.
Three hour lecture

ARCH 4885 [0.5 credit]

Building Pathology and Rehabilitation

Deterioration mechanisms for concrete, timber, steel and masonry structures. Identification of design deficiencies; criteria for selection and design of rehabilitation systems. Design techniques to reduce deterioration in new construction and historical structures. Includes: Experiential Learning Activity
Also listed as ACSE 4601, ARCN 4200, CIVE 4601.
Prerequisite(s): ARCH 3881.
Lectures three hours a week, lab/field work two hours a week.

ARCH 4991 [0.5 credit]

History of Modern Housing

Study of housing as a function of social organization, demographics, market demand and public policy. Topics include the evolution of housing form, the role of the state, and the participation of architects in the housing marketplace in the 19th and 20th century. Precludes additional credit for ARCH 4201.
Prerequisite(s): Third year standing in the Bachelor of Architectural Studies (BAS) program.
Three hour lecture

ARCH 4992 [0.5 credit]

Theories of Urbanism

Contemporary urban theory and critical scholarship that engages evolving social, political, economic and environmental perspectives, addresses multiple scales, geographic contexts, and disciplinary boundaries, and investigates the expanding array of models, tools and techniques that have contributed to various theories of urbanism. Precludes additional credit for ARCU 4300.
Prerequisite(s): Third year standing in the Bachelor of Architectural Studies (BAS) program.
Three hour lecture

ARCH 4993 [0.5 credit]

Topics in Urbanism

Advanced seminar in selected topics related to urbanism. Topics may include histories and theories related to urban systems, design, and planning. Precludes additional credit for ARCU 4801.
Prerequisite(s): Third year standing in the Bachelor of Architectural Studies (BAS) Urbanism stream.
Three hour seminar

Architecture (ARCU) Courses

ARCU 3100 [0.5 credit]

The Morphology of the City

Primary structural, spatial and formal organization and elements that characterize the morphology of cities; historical and contemporary significance for architecture and urban design. (Core). Precludes additional credit for ARCH 2331.
Prerequisite(s): First-year standing in the B.A.S. Urbanism major, second or third-year standing in other B.A.S. majors, or permission of the School.
Lecture two hours a week and tutorial one hour a week.

ARCU 4103 [0.5 credit]**Cities**

Course addresses cities such as Istanbul, Mexico City, Venice, Paris, Ottawa, Mumbai, and New Orleans. Topics presented by the instructor and guests include environmental resilience and climate change; social justice and informal settlement; smart cities and data privacy; and urban design, memory, and imagination.

Precludes additional credit for ARCU 3902 (no longer offered).

Prerequisite(s): Second-year standing or permission of the Instructor.

Lecture two hours per week and tutorial one hour per week.

ARCU 4300 [0.5 credit]**Theories of Urbanism**

Contemporary urban theory and critical scholarship that engages evolving social, political, economic and environmental perspectives, addresses multiple scales, geographic contexts, and disciplinary boundaries, and investigates the expanding array of models, tools and techniques that have contributed to various theories of urbanism.

Precludes additional credit for ARCH 4992.

Prerequisite(s): ARCU 3100.

ARCU 4700 [0.5 credit]**Urban Utopias**

Urban utopias throughout history, with emphasis on the 20th century. Garden Cities, anti-urbanism and radical decentralization, the city in the region, Italian Rationalist cities, Le Corbusier and CIAM, post-WWII New Towns (England, Scandinavia and the US), Sustainable Urbanism.

Prerequisite(s): third or fourth-year standing in B.A.S.

Urbanism program or permission of the School.

Lectures three hours a week.

ARCU 4801 [0.5 credit]**Topics in Urbanism**

Advanced seminar in selected topics related to urbanism.

Topics may include histories and theories related to urban systems, design, and planning. (Core course).

Precludes additional credit for ARCH 4993.

Prerequisite(s): third-year standing in B.A.S. (Urbanism) or permission of the Instructor.

Seminar three hours per week.